

PROBLEM STATEMENT 26189

# AI-Powered Criminal Network Analysis System

*Uncovering hidden criminal networks from fragmented, unstructured case data — entity extraction, relationship mapping, and influence analysis for investigators.*

|              |                                                              |
|--------------|--------------------------------------------------------------|
| Organization | Ministry of Home Affairs                                     |
| Department   | National Crime Records Bureau (NCRB) — Women Safety Division |
| Category     | Software                                                     |
| Theme        | Blockchain & Cybersecurity                                   |



## THE PROBLEM

### Why Manual Analysis Falls Short

Criminal activity today is organized and interconnected — associates, intermediaries, financial channels, communication links, locations, and events all form part of a single hidden network. Investigators draw this picture from FIRs and police reports, call detail records, financial transactions, surveillance reports, social media intelligence, criminal history databases, and intelligence reports.

That data is fragmented, unstructured, and scattered across systems — which makes manual analysis slow, labour-intensive, and prone to missing the connections that matter most.

### What the system needs to do

- Collect and process data from multiple sources
- Extract key entities — people, locations, vehicles, phone numbers, organizations
- Build relationship maps showing how entities are connected
- Identify key individuals who play influential roles in the network
- Detect suspicious patterns and unusual activity
- Provide visual and analytical insight for investigators

*Expected outcome: an AI-powered system that automatically analyses structured and unstructured crime-related data to uncover criminal networks, identify key influencers, detect suspicious patterns, and deliver actionable intelligence.*

## REFERENCE

### Tools at a Glance

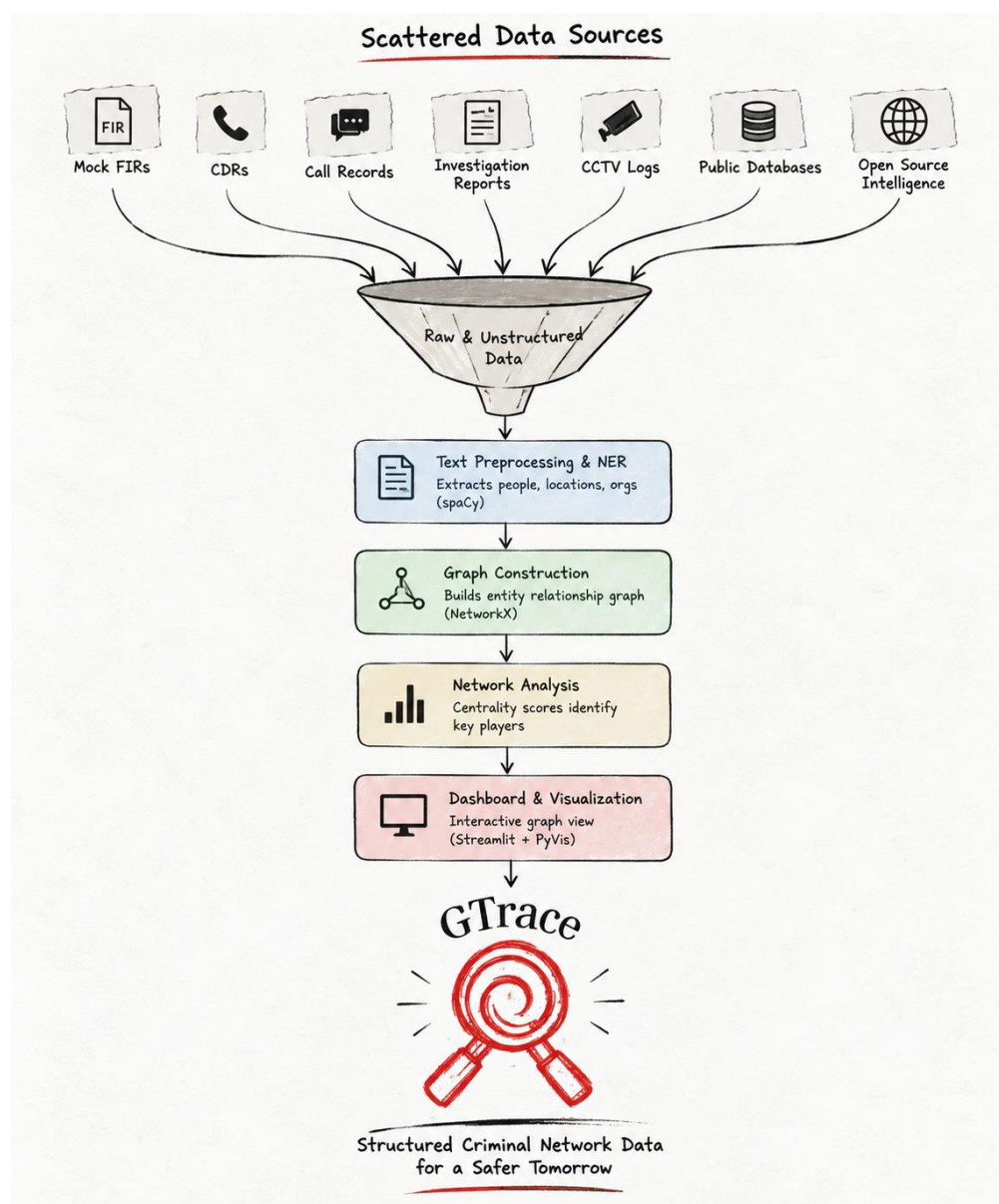
| Purpose                | Tool                   | Why                                                                  |
|------------------------|------------------------|----------------------------------------------------------------------|
| Mock data generation   | Faker                  | Realistic fake names, locations, phone numbers — no real data needed |
| Entity extraction      | spaCy (en_core_web_sm) | Pretrained NER, no training required                                 |
| Custom entity patterns | Regex                  | Catches phone numbers and plates spaCy misses                        |
| Graph building         | NetworkX               | Mature graph library with built-in centrality algorithms             |
| Graph visualization    | PyVis                  | Interactive, embeddable network graphs                               |
| Dashboard / app        | Streamlit              | Fast way to wrap everything in a usable interface                    |

## OUR APPROACH

### A Five-Stage Pipeline

Every stage rests on a mature, pretrained tool — nothing in the pipeline needs to be trained from scratch. That keeps the build achievable for a first hackathon while still producing a genuinely working system, end to end.

1. Data sources — mock FIRs, CDRs, call records, and reports
2. Text preprocessing & NER — extract people, locations, organizations (spaCy)
3. Graph construction — build the entity relationship graph (NetworkX)
4. Network analysis — centrality scores surface the key players
5. Dashboard & visualization — an interactive graph view (Streamlit + PyVis)



## STAGE 01 – 02

## Mock Data & Entity Extraction

### Building a realistic mock dataset

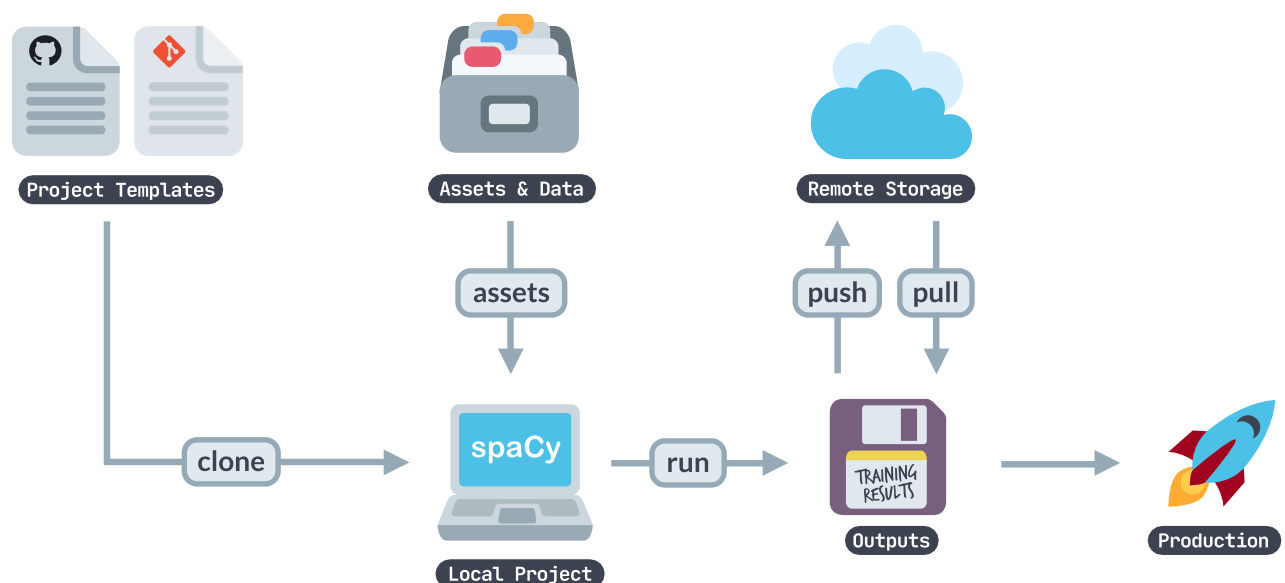
Real criminal records aren't accessible for a hackathon, so the dataset is synthetic by design. The Faker library generates realistic fake names, cities, and phone numbers — using the Indian locale so the data feels grounded.

Around 15–20 short mock FIR or report snippets are hand-written, with deliberately overlapping names, shared phone numbers, and shared locations, so the resulting graph has real structure to uncover. The cleanest way in is to sketch a small network on paper first — one kingpin, two lieutenants, a handful of associates, a few red herrings — and write the documents around that shape.

### Extracting entities with spaCy

spaCy's small pretrained English model (en\_core\_web\_sm) needs no training — it's loaded and run directly on each mock document, and entities are pulled from doc.ents as (text, label) pairs: PERSON, GPE for locations, ORG for organizations.

spaCy won't catch phone numbers or vehicle plates on its own, so simple regex patterns run alongside it to fill that gap, rather than training a custom NER model.



## Connecting people

Shared attributes add real signal on top — people who share a phone number in a call record, appear at the same location and time, or link through the same vehicle or organization. This carries far more meaning than co-occurrence alone, and combining the two is enough for a strong, demoable graph.

## Finding the key players

**Vikram Singh**

| Type       | Person       |
|------------|--------------|
| Centrality | 0.87         |
| Aliases    | V. Singh, VS |
| Linked To  | 12 entities  |

**Key Relations**

- Sameer Khan (Operations)
- Imran Qureshi (Finance)
- Shakti Group (Organization)
- Mumbai (Location)
- FIR-1023 (Event)

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## FEASIBILITY

## Future Scope

1. **Blockchain Audit Log** — Store investigation actions and important graph updates on a tamper-evident blockchain ledger for traceability and accountability.
2. **Multi-agency Data Sharing** — Enable secure sharing of relevant intelligence between authorized agencies while maintaining access controls.
3. **Scalable Graph Database** — Move from in-memory NetworkX graphs to systems such as Neo4j for handling much larger criminal networks.
4. **Real-time Data Integration** — Connect with authorized CDR, FIR, case-management, and other law-enforcement data sources instead of mock datasets.

## Challenges & How We Tackle Them

| Challenge                                                                                                     | How we tackle it                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Messy and inconsistent data</b> — typos, missing fields, different formats, regional language              | <b>Data cleaning + normalization</b> before NLP; standardize names, phone numbers, locations and dates.                                        |
| <b>False relationships</b> — two people appearing in the same FIR doesn't necessarily mean they are connected | <b>Relationship confidence scoring</b> based on the type and frequency of evidence instead of creating a strong edge from a single mention.    |
| <b>Same name, different person</b> — e.g., multiple people named “Ravi Kumar”                                 | <b>Entity resolution</b> using additional attributes such as phone number, location, organization and case information.                        |
| <b>Indirect/hidden connections are difficult to detect</b>                                                    | Use <b>multi-hop graph traversal and centrality/community analysis</b> to identify connections beyond direct relationships.                    |
| <b>Limited labelled crime data for ML models</b>                                                              | Start with <b>rule-based + graph-based analysis</b> , then improve models as verified investigation data becomes available.                    |
| <b>False positives in key-player detection</b>                                                                | Combine <b>multiple network metrics</b> (degree, betweenness, PageRank/centrality) rather than relying on a single score.                      |
| <b>Independent criminals may have no network connections</b>                                                  | Complement graph analysis with <b>anomaly detection and individual-level indicators</b> , rather than depending entirely on network structure. |
| <b>Data privacy and unauthorized access</b>                                                                   | Implement <b>role-based access control, encryption and audit logging</b> ; future versions can add the proposed blockchain-based audit trail.  |